

PDC Reconstruction Literature Survey and SAMURAI Terminal Air-MS Gap Analysis

Baiting Tian
RIKEN Nishina Center / DPOL Experiment

2026-04-26

Contents

1	TL;DR	3
2	The problem: air multiple scattering at the SAMURAI terminal	5
2.1	Geometry of the proton flight path	5
2.2	Measured σ values	5
2.3	Gap to design specifications	5
2.4	Why this matters for the science floor	5
3	PDC reconstruction reference frame	5
3.1	Hardware foundations	5
3.2	Algorithm spine	5
3.3	Performance achieved across SAMURAI campaigns	5
4	Gap analysis	5
4.1	Performance: σ values	5
4.2	Methodology	5
4.3	Material budget and vacuum configuration	5
4.4	Experimental configuration and geometry	5
5	Broader MINOS+SEASTAR chain context	5
5.1	SEASTAR programme timeline	5
5.2	Companion detectors	5
5.3	Reviews and resources	5
5.4	Caveats from the broader bibliography	5
6	Implications and next steps	5
6.1	Short-term data acquisition action items	5
6.2	Medium-term simulation and analysis directions	5
6.3	Open questions	5
6.4	Non-goals	5
7	References	5
7.1	Hardware and methodology (peer-reviewed, foundational)	5
7.2	Algorithm references	5
7.3	PhD theses with PDC methodology	5
7.4	Physics papers using PDC1/PDC2 (confirmed/implicit)	5
7.5	Excluded / context-only	5
7.6	Reviews and topical collections	5

7.7	Construction proposals and internal notes	5
7.8	Project-internal references	5

1 TL;DR

This document surveys the published reference frame for the SAMURAI PDC1/PDC2 drift-chamber chain and contrasts it with our current Geant4 reconstruction performance, in order to localise the air multiple-scattering gap that drives our PDC2 spatial resolution one order of magnitude above design.

- **Reference frame.** The PDC1/PDC2 chain has no dedicated peer-reviewed publication. The canonical references are the SAMURAI overview paper by Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013) §3.4¹, the RIKEN internal technical note *Detector-PDC.pdf*², and the Kahlbow PhD thesis (TU Darmstadt, 2019)³ which documents an independent reconstruction of the same hardware.
- **Design specifications.** Combining Kobayashi 2013 §3.4¹ with the SAMURAI 2012 construction proposal⁴, the PDC1/PDC2 chain targets 1 mm position resolution, 2 mrad angular resolution, and 0.8% relative momentum resolution ($\Delta p/p$). Commissioning RI-beam runs reached $\Delta p/p \approx 1/1500$ ¹; the original proton-mode design value is $\Delta p/p \approx 1/700$ as quoted in the RIBF-TAC05 (2005) construction proposal⁵.
- **Our measured $\sigma_y(\text{PDC2})$.** In our Geant4 + reconstruction chain we observe $\sigma_y(\text{PDC2}) = 12.83\text{--}14.93$ mm in the all-air configuration, 6.80–8.21 mm when Region A (dipole cavity + beam pipe) is set to vacuum (`beamline_vacuum`), and approximately 1.0 mm in the all-vacuum configuration⁶. The realistic-air values sit roughly an order of magnitude away from the 1 mm hardware design specification¹.
- **Air MS dominates and the configuration choice matters.** A quadrature decomposition⁶ attributes the all-air budget to $\sigma_A \approx 11.19$ mm from Region A (cavity + beam pipe, ~ 4.3 m) and $\sigma_B \approx 7.43$ mm from Region B (Exit Window $\rightarrow$ PDCs, ~ 2 m). The stage D inputs from the same memo⁷ give $\sigma_y(\text{PDC2}) = 7.49$ mm if Region A is vacuum and $\sigma_y(\text{PDC2}) = 13.50$ mm if both Region A and Region B are air. Whether Region A is air or vacuum during the experiment is therefore a configuration question that selects between these two values; this document does not assert which one corresponds to the actual run.

See §6 for proposed action items.

¹T. Kobayashi et al., Nucl. Instr. Meth. B **317**, 294–304 (2013), DOI: [10.1016/j.nimb.2013.05.089](https://doi.org/10.1016/j.nimb.2013.05.089).

²RIKEN Nishina Center, *Detector-PDC* technical note, <https://www.nishina.riken.jp/ribf/SAMURAI/image/Detector-PDC.pdf>.

³J. Kahlbow, PhD thesis, TU Darmstadt (2019), TUPrints 9192, <https://tuprints.ulb.tu-darmstadt.de/9192/>.

⁴SAMURAI Collaboration, *SAMURAI Construction Proposal* (2012), https://ribf.riken.jp/SAMURAI/120425_SAMURAIConstProp.pdf.

⁵SAMURAI Collaboration, *RIBF-TAC05 SAMURAI construction proposal* (2005), https://ribf.riken.jp/RIBF-TAC05/10_SAMURAI.pdf.

⁶See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §4.

⁷See `docs/reports/reconstruction/momentum_reco/ms_ablation/ms_ablation_mixed_20260422_zh.md` §6.

2 The problem: air multiple scattering at the SAMURAI terminal

2.1 Geometry of the proton flight path

2.2 Measured σ values

2.3 Gap to design specifications

2.4 Why this matters for the science floor

3 PDC reconstruction reference frame

3.1 Hardware foundations

3.2 Algorithm spine

3.3 Performance achieved across SAMURAI campaigns

4 Gap analysis

4.1 Performance: σ values

4.2 Methodology

4.3 Material budget and vacuum configuration

4.4 Experimental configuration and geometry

5 Broader MINOS+SEASTAR chain context

5.1 SEASTAR programme timeline

5.2 Companion detectors

5.3 Reviews and resources

5.4 Caveats from the broader bibliography

6 Implications and next steps

6.1 Short-term data acquisition action items

6.2 Medium-term simulation and analysis directions

6.3 Open questions

6.4 Non-goals

7 References

7.1 Hardware and methodology (peer-reviewed, foundational)

7.2 Algorithm references

7.3 PhD theses with PDC methodology

7.4 Physics papers using PDC1/PDC2 (confirmed/implicit)

7.5 Excluded / context-only

7.6 Reviews and topical collections

7.7 Construction proposals and internal notes

7.8 Project-internal references